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Forecasting agricultural commodity prices using machine learning models

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Abstract

Accurate forecasting of agricultural commodity prices is important for market planning, risk assessment, and agricultural policy support. This study develops a historical-price-based forecasting framework for nine agricultural commodities in Maharashtra, India, using mandi-level AGMARKNET data from 2001 to October 2025. Daily market records were cleaned and aggregated into monthly modal prices, and lag, rolling-window, and seasonal calendar features were used to generate one-step-ahead monthly forecasts for January 2021–October 2025. To avoid temporal leakage, outlier limits and preprocessing operations required for modelling were estimated from the available training history within each sequential forecasting step. Random Forest, Extra Trees, Histogram Gradient Boosting, and XGBoost models were compared with one-month Naive and 12-month Seasonal Naive benchmarks. The benchmark evaluation shows that the one-month Naive forecast achieved the lowest MAPE for all commodities, with MAPE values ranging from 2.30% for Paddy to 4.91% for Sunflower. The best machine learning models consistently improved over the Seasonal Naive benchmark but did not outperform the one-month Naive benchmark, indicating strong short-run price persistence in the monthly series. These findings provide a transparent and leakage-safe assessment of historical-price-only machine learning models for agricultural commodity price forecasting and highlight the need for future integration of exogenous market, production, weather, and policy variables.

Keywords Agricultural commodity, Price forecasting, Machine learning, Time-series forecasting, Agricultural markets, Feature engineering, Decision support systems

1 Introduction

Agriculture plays a critical role in the economy of many developing countries, particularly India, where a substantial proportion of the population depends on agricultural activities for livelihood. Fluctuations in agricultural commodity prices directly influence farmers' income, market stability, policy planning, and food security. Accurate forecasting of commodity prices enables stakeholders, including farmers, traders, policymakers, and agricultural organisations, to make informed decisions regarding production, storage, marketing, and resource allocation.

Traditional forecasting approaches such as statistical time-series models have been widely applied to agricultural markets. However, the increasing availability of large-scale

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agricultural datasets and advances in computational techniques have encouraged the adoption of machine learning methods for forecasting applications. Recent studies have demonstrated that machine learning (ML) algorithms can effectively capture nonlinear relationships and complex temporal patterns present in agricultural commodity markets [1–3].

The integration of artificial intelligence and ML into agricultural systems has accelerated the development of intelligent forecasting frameworks. Ensemble learning techniques, including Random Forest, Gradient Boosting, XGBoost, and Extra Trees, have shown promising results in agricultural prediction tasks due to their ability to model nonlinear interactions and reduce forecasting errors [4–7]. Recent research has focused on agricultural commodity forecasting using machine learning and hybrid approaches. Several studies have investigated crop recommendation systems, crop yield prediction, agricultural market forecasting, and commodity price prediction under varying climatic and economic conditions [8, 9]. In addition, comprehensive reviews have highlighted the growing importance of machine learning, deep learning, and hybrid forecasting models in agricultural applications [2, 10].

Several researchers have emphasised the importance of feature engineering, external market factors, weather variables, and temporal characteristics in improving forecasting accuracy [2, 3]. Hybrid and knowledge-informed machine learning frameworks have also demonstrated enhanced predictive performance for agricultural forecasting tasks [1, 2]. The emergence of interpretable ensemble learning techniques further supports the deployment of machine learning systems in real-world agricultural environments [4, 6].

Despite these advancements, accurate forecasting of agricultural commodity prices remains challenging due to seasonality, market volatility, weather variability, and changing economic conditions. Therefore, there is a need for robust forecasting frameworks that evaluate historical market information under realistic temporal validation [2, 10].

Motivated by these challenges, the present study develops a ML-based forecasting framework for predicting agricultural commodity prices in Maharashtra, India. Historical commodity market data are processed through a comprehensive feature engineering pipeline, and multiple ensemble learning algorithms are evaluated using standard forecasting performance metrics. The proposed framework aims to identify the most suitable predictive model for major agricultural commodities and provide insights into their market behaviour.

2 Literature review

Agricultural commodity price forecasting has attracted considerable attention due to its importance in supporting market planning, risk management, and policy formulation. Traditional forecasting methods have gradually evolved into ML and hybrid forecasting frameworks capable of handling large-scale and complex agricultural datasets.

Oktoviany et al. [3] proposed a ML-based framework for agricultural commodity price state prediction by incorporating external economic and environmental factors. Their study demonstrated that integrating external information significantly improves forecasting performance. Sari et al. [1] investigated various optimised machine learning techniques for agricultural commodity price forecasting and reported superior predictive accuracy compared to conventional statistical approaches.

Sun et al. [2] summarised recent developments in agricultural product price forecasting and highlighted the increasing adoption of machine learning, deep learning, and hybrid methods. Tran et al. [10] further reviewed machine learning applications in agricultural commodity markets and emphasised the importance of advanced forecasting techniques for improving decision-making processes.

Several studies have focused on machine learning applications in agriculture beyond price forecasting. Mahale et al. [8] developed a machine learning-based crop recommendation and forecasting system for Maharashtra using LSTM networks. Nagesh et al. [9] employed boosting-based machine learning techniques for crop yield prediction. These studies indicate the broader use of machine learning in agricultural prediction tasks, although their objectives differ from monthly commodity price forecasting.

Feature engineering has emerged as a critical component of forecasting systems. Reviews of agricultural price forecasting emphasise that lagged values, temporal aggregation, seasonal variables, and suitable benchmark comparisons are necessary for evaluating model performance in time-series settings [2, 10, 11].

Tree-based ensemble methods continue to dominate forecasting applications due to their strong predictive capabilities. Random Forest [4], Gradient Boosting Machines [5], Extra Trees [7], and XGBoost [6] remain among the most widely adopted machine learning algorithms for regression and forecasting tasks.

Although previous studies have reported encouraging results, limited research has systematically compared multiple ensemble learning algorithms for forecasting major agricultural commodities using a unified framework and extensive historical market data. Therefore, the present study addresses this gap by evaluating several machine learning models on long-term commodity price data from Maharashtra and identifying the most effective forecasting approach.

2.1 Research gap and objectives

Although substantial research has been conducted on agricultural commodity price forecasting using statistical, ML, and deep learning approaches, several limitations remain. Many existing studies focus on a single commodity, utilise relatively short time periods, or depend heavily on external explanatory variables such as weather conditions, crop arrivals, macroeconomic indicators, and international market information. Such dependencies may limit the applicability of forecasting models in regions where external data are unavailable, inconsistent, or difficult to obtain.

Furthermore, limited attention has been given to developing a unified forecasting framework capable of handling multiple agricultural commodities using long-term historical market price data while preserving temporal dependencies through realistic validation strategies. To address these gaps, the present study develops a ML-based forecasting framework for nine major agricultural commodities using mandi-level price data from 2001 to October 2025. Specifically, the study seeks to preprocess and transform historical price data, generate lag-based and seasonal features, implement a sequential expanding-window one-step-ahead monthly forecasting protocol, and evaluate Random Forest, Extra Trees, Histogram Gradient Boosting, and XGBoost models against Naive and Seasonal Naive benchmarks. The objective is to provide a reproducible assessment of what can and cannot be achieved using only historical price information.

3 Methodology

3.1 Data description

The dataset contains daily mandi-level price records for nine agricultural commodities: Arhar, Bajra, Cotton, Maize, Paddy, Sesamum, Soyabean, Sunflower, and Wheat. Each record includes date, commodity name, market information,district, state, variety, minimum price, maximum price, and modal price. The modal price (p_{modal}) was used as the target variable because it represents the most commonly observed market price for a commodity on a particular date.

Table 1 summarizes the statistical characteristics of the commodity price dataset used in this study. Considerable differences can be observed in the price distributions, average market prices, and variability among commodities, reflecting the heterogeneous nature of agricultural markets. These results highlight the presence of diverse pricing behaviour and justify the need for commodity-specific forecasting models within a unified machine learning framework.

3.2 Proposed workflow

The overall workflow of the study consists of data collection, pre-processing, monthly aggregation, feature engineering, model training, benchmark comparison, sequential expanding-window validation, evaluation, and result interpretation. Figure 1 illustrates the proposed machine learning framework for agricultural commodity price forecasting. Historical mandi price data and calendar information are processed to create commodity-wise monthly series. After preprocessing and feature engineering, fixed-configuration machine learning models and Naive benchmarks are evaluated under the same one-step-ahead monthly forecasting protocol.

3.3 Data preprocessing

Data preprocessing was performed to improve data quality and ensure suitability for forecasting. The date column was converted into datetime format, duplicate records were removed, and invalid or missing dates were excluded. Price variables were converted into numerical format. Since modal price is the target variable, records with missing modal price values were removed. Missing values in the target variable were not imputed because artificial target generation may introduce bias into supervised learning. Records with missing modal prices were therefore excluded, while invalid and extreme price values were handled through filtering and commodity-wise outlier capping. Non-positive prices were treated as invalid.

Table 1 Data description by commodity

Commodity	Obs.	Min	Max	Mean	Median	Std. Dev.	CV (%)
Arhar	300,560	1100.00	12380.00	4834.86	4400.00	2299.21	47.55
Bajra	198,722	401.00	3900.00	1417.30	1325.00	654.38	46.17
Cotton	94,900	1325.00	12006.00	4977.14	4834.00	1769.81	35.56
Maize	186,645	400.00	4000.00	1344.01	1300.00	560.74	41.72
Paddy	78,090	381.36	3449.11	1451.23	1400.00	691.74	47.67
Sesamum	27,832	1440.00	21000.00	6996.91	6303.00	3797.26	54.27
Soyabean	336,095	981.00	9099.06	3376.36	3376.00	1397.85	41.40
Sunflower	35,930	972.43	7100.00	2520.75	2150.00	1159.38	45.99
Wheat	395,294	650.00	4800.00	1746.25	1675.00	614.59	35.19

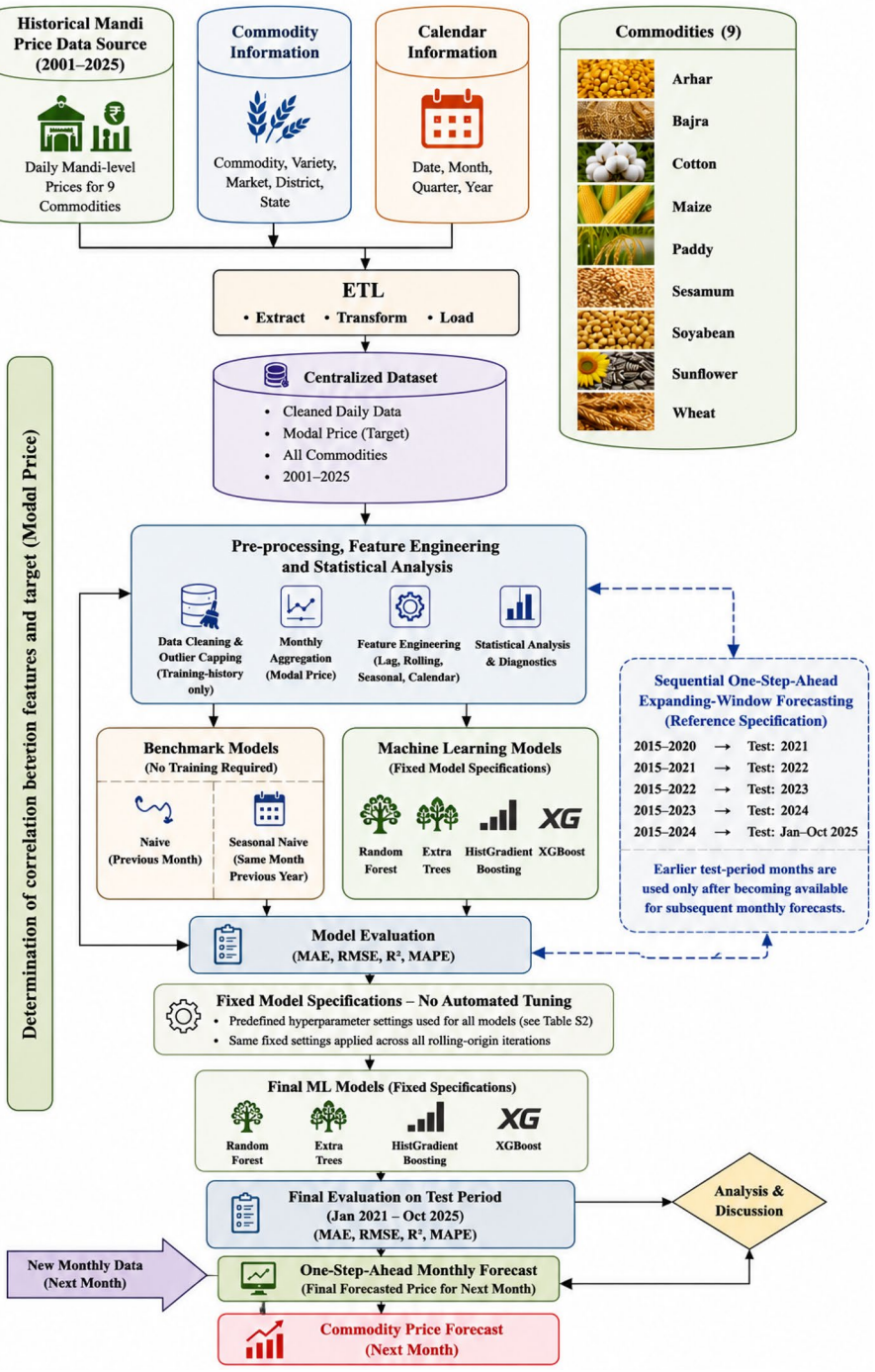


Fig. 1 Proposed workflow for agricultural commodity price forecasting

Extreme values were handled through commodity-wise outlier capping using the 0.1st and 99.9th percentiles. In the forecasting experiment, these quantile limits were estimated only from the training history available at each forecasting step and then applied to the corresponding test month. This prevents future test-period information from influencing preprocessing. For each commodity c, the capped price value is defined as:

$$P'_{i,c} = \begin{cases} Q_{0.001,c}, & \text{if } P_{i,c} < Q_{0.001,c} \\ P_{i,c}, & \text{if } Q_{0.001,c} \leq P_{i,c} \leq Q_{0.999,c} \\ Q_{0.999,c}, & \text{if } P_{i,c} > Q_{0.999,c} \end{cases} \quad (1)$$

where $P_{i,c}$ is the original price, $P'_{i,c}$ is the capped price, and $Q_{0.001,c}$ and $Q_{0.999,c}$ represent the lower and upper commodity-specific quantiles.

3.4 Log transformation

Agricultural price series often show right-skewness because of seasonal effects, demand shocks, and sudden price increases. Therefore, logarithmic transformation was applied to stabilise variance:

$$y_t = \log(1 + P_t) \quad (2)$$

where P_t denotes the monthly modal price at time t, and y_t represents the log-transformed price. Predicted values were converted back to the original scale as:

$$\hat{P}_t = e^{\hat{y}_t} - 1 \quad (3)$$

where $\hat{y}_t$ is the predicted log-transformed value and $\hat{P}_t$ is the corresponding predicted modal price on the original scale. Direct inverse transformation from the log scale may introduce retransformation bias. No explicit retransformation-bias correction was applied in the present analysis, and this is acknowledged as a methodological limitation and potential direction for future work.

Figure 2 presents the distribution of log-transformed modal prices for the selected agricultural commodities. The boxplots indicate that the logarithmic transformation effectively reduces skewness and compresses extreme price values, resulting in more stable and comparable distributions across commodities. This transformation improves the suitability of the data for machine learning models by reducing the influence of large price variations and enhancing numerical stability during model training.

3.5 Monthly aggregation

The daily mandi-level price records were obtained from the AGMARKNET portal maintained by the Directorate of Marketing and Inspection, Government of India [12].

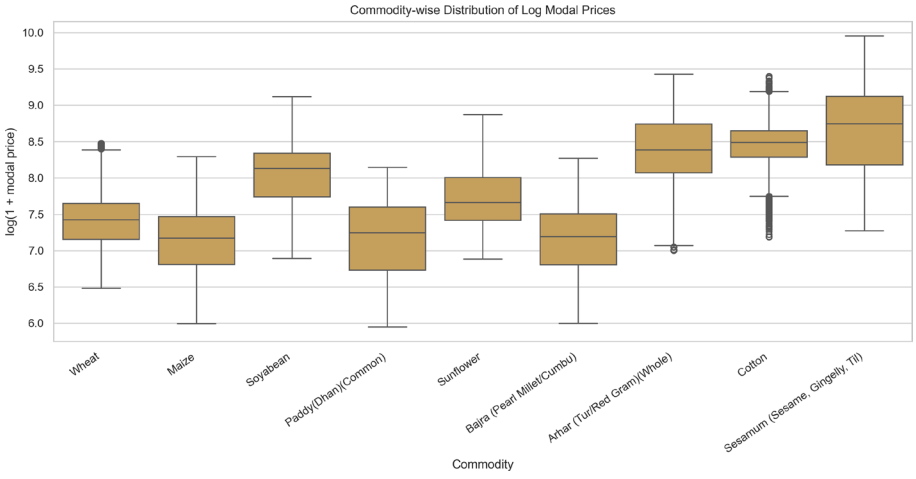


Fig. 2 Commodity-wise distribution of log modal prices

For forecasting, records were aggregated into monthly average modal prices to reduce daily market noise and irregular market reporting. The forecasting unit is therefore the commodity-month, not the individual daily mandi record. The dataset was restricted to Maharashtra; observations from reporting mandis and varieties were pooled into one commodity-level monthly series with equal weight for each valid mandi-level observation. Reporting-count diagnostics and mean-versus-median robustness checks are provided in the Supplementary Material. For commodity c in month m, the monthly average modal price was calculated as:

$$\bar{P}_{m,c} = \frac{1}{n_{m,c}} \sum_{i=1}^{n_{m,c}} P_{i,m,c} \tag{4}$$

where $\bar{P}_{m,c}$ denotes the monthly average modal price of commodity c during month m, $P_{i,m,c}$ represents the modal price of the i-th observation, and $n_{m,c}$ denotes the total number of observations available for commodity c in month m.

Figure 3 represents monthly trends; it is observed that most commodities experienced long-term upward movement in modal prices. However, the pattern is not uniform across commodities. Some commodities show relatively smooth growth, while others display sudden spikes and corrections. Such differences explain why a single machine learning model does not perform equally well for all commodities.

3.6 Feature engineering

Feature engineering was carried out to convert the monthly commodity price series into meaningful explanatory variables for the forecasting models. Since the current price of an agricultural commodity is influenced by its past prices, short-term movements, long-term trends, volatility, and seasonal effects, lag-based, rolling-window, and calendar-based features were constructed [2].

3.6.1 Lag features

Lag features were generated to include previous price information in the model. For a monthly modal price series P_t, the lag feature of order k at time t is defined as:

$$\text{Lag}_{k,t} = P_{t-k} \tag{5}$$

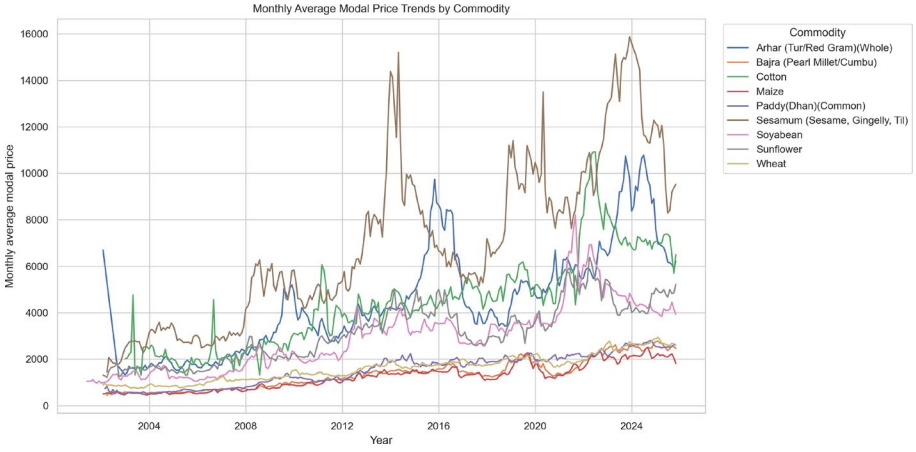


Fig. 3 Monthly average modal price trends by commodity

where P_t denotes the modal price at month t, and P_{t-k} denotes the price observed k months before month t. In this study, lag values of $k = 1, 2, 3, 6, 9, 12, 18$, and 24 months were used. The short-term lags of 1, 2, and 3 months capture recent price movements, while the longer lags of 6, 9, 12, 18, and 24 months capture medium-term and annual price behaviour.

3.6.2 Rolling mean features

Rolling mean features were computed to capture the average price trend over previous months. For a rolling window of size w, the rolling mean at time t is defined as:

$$\mathrm{RM}_{w,t} = \frac{1}{w} \sum_{j=1}^{w} P_{t-j} \tag{6}$$

where w represents the number of previous months considered in the rolling window. Rolling windows of $w = 3, 6, 12$, and 24 months were used. The 3-month rolling mean captures short-term price trend, the 6-month rolling mean captures medium-term movement, the 12-month rolling mean represents annual price behaviour, and the 24-month rolling mean captures long-term market trend. These features help smooth irregular price fluctuations and provide stable trend information to the forecasting models.

3.6.3 Rolling standard deviation features

Rolling standard deviation features were used to measure price volatility over different historical windows. For a rolling window of size w, the rolling standard deviation at time t is calculated as:

$$\mathrm{RSD}_{w,t} = \sqrt{\frac{1}{w-1} \sum_{j=1}^{w} \left(P_{t-j} - \mathrm{RM}_{w,t}\right)^2} \tag{7}$$

where $\mathrm{RM}_{w,t}$ is the rolling mean calculated over the same window size w. The window sizes $w = 3, 6, 12$, and 24 months were also used for rolling standard deviation. The 3-month rolling standard deviation captures short-term volatility, the 6-month rolling standard deviation captures medium-term market variation, the 12-month rolling standard deviation reflects annual volatility, and the 24-month rolling standard deviation captures long-term price instability. These variables help the model identify periods of stable and fluctuating commodity prices.

3.6.4 Calendar and seasonal features

Calendar-based features were also included because agricultural commodity prices are affected by seasonal production cycles, harvesting periods, supply availability, and market demand. Month, quarter, and year index were added as explanatory variables. To represent annual seasonality more effectively, sine and cosine transformations of the month number were generated.

The sine transformation of month is defined as:

$$\mathrm{Month}_{\sin,t} = \sin\left(\frac{2\pi m_t}{12}\right) \tag{8}$$

The cosine transformation of month is defined as:

$$\text{Month}_{\cos,t} = \cos\left(\frac{2\pi m_t}{12}\right) \tag{9}$$

where m_t denotes the month number corresponding to time t, with $m_t = 1, 2, \ldots, 12$. These transformations preserve the cyclical nature of months, because December and January are treated as neighbouring months instead of distant numerical values. Thus, the final feature set included lag features, rolling mean features, rolling volatility features, and seasonal calendar features for forecasting agricultural commodity prices.

Feature engineering plays a crucial role in enhancing the predictive capability of machine learning models for commodity price forecasting. As illustrated in Fig. 4, a systematic feature engineering workflow was developed to generate lag-based, rolling-window, and seasonal features from the monthly modal price series. These engineered variables were designed to capture temporal dependencies, market trends, price volatility, and cyclical seasonal patterns present in agricultural commodity prices.

Algorithm 1 Feature Engineering for Commodity Price Forecasting

Require: Monthly modal price series $P = \{P_1, P_2, \ldots, P_T\}$
Ensure: Feature matrix X and target vector y

Initialise empty dataset $\mathcal{D}$
for $t = 25$ **to** T **do**
  Generate lag features: $L_k(t) = P_{t-k}, \quad k \in \{1, 2, 3, 6, 9, 12, 18, 24\}$
  Generate rolling mean features: $RM_w(t) = \frac{1}{w}\sum_{j=1}^{w} P_{t-j}, \quad w \in \{3, 6, 12, 24\}$
  Generate rolling std features: $RSD_w(t) = \sqrt{\frac{1}{w-1}\sum_{j=1}^{w}(P_{t-j} - RM_w(t))^2}$
  Generate seasonal features: $S_{\sin}(t) = \sin\left(\frac{2\pi m_t}{12}\right), \quad S_{\cos}(t) = \cos\left(\frac{2\pi m_t}{12}\right)$
  Construct feature vector: $X_t = \{L_k(t), RM_w(t), RSD_w(t), S_{\sin}(t), S_{\cos}(t)\}$
  Set target: $y_t = P_t$
  Append (X_t, y_t) to $\mathcal{D}$
end for
return X, y

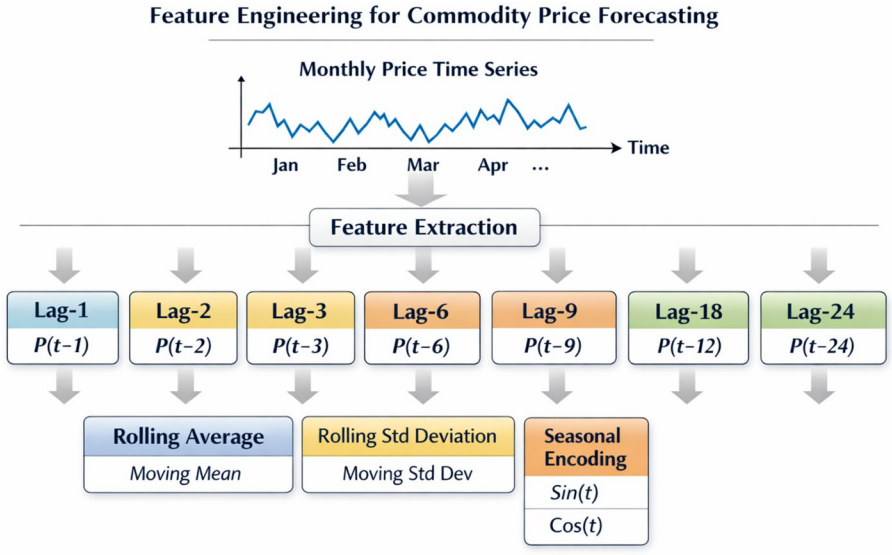


Fig. 4 Feature engineering workflow for agricultural commodity price forecasting using lag-based, rolling-window, and seasonal features

3.7 Training and testing strategy

To preserve the temporal structure of the dataset, random splitting is not used. Instead, the analysis uses sequential one-step-ahead monthly forecasting for the test period January 2021–October 2025. For each test month t, all features are constructed only from information available before month t, and the target is the modal price for month t. Actual prices observed earlier in the test period are therefore used only after they become available. For example, the January 2021 forecast uses lagged information available up to December 2020, whereas the September 2021 forecast may use observed January–August 2021 prices to construct lag and rolling-window predictors.

The main specification uses 2015 as the initial training start year. The 2015 start year was retained as the predefined reference specification, while sensitivity analyses using 2001 and 2010 were additionally conducted to assess dependence on training-history length and are reported in the Supplementary Material (Table 2).

For a test month t, the training and testing sets are:

$$D^t_{\text{train}} = \{(X_\tau, y_\tau) : \tau < t\} \tag{10}$$

$$D^t_{\text{test}} = \{(X_t, y_t)\} \tag{11}$$

where X_t is the feature vector and y_t is the log-transformed modal price. This design evaluates realistic one-step-ahead monthly forecasting, not random holdout prediction or full-year-ahead forecasting from the previous December.

Two simple benchmark models were also evaluated. The one-month Naive forecast is:

$$\hat{P}^{\text{Naive}}_t = P_{t-1} \tag{12}$$

and the 12-month Seasonal Naive forecast is:

$$\hat{P}^{\text{SNaive}}_t = P_{t-12}. \tag{13}$$

These benchmarks provide a necessary reference point for judging whether machine learning models improve on simple price persistence and annual seasonality.

3.8 Machine learning models

Four machine learning regression models were evaluated: Random Forest, Extra Trees, Histogram Gradient Boosting, and XGBoost [6]. The models used fixed hyperparameter settings rather than automated hyperparameter tuning; the exact settings are reported in the Supplementary Material. Random Forest combines multiple decision trees trained on bootstrapped samples. Extra Trees is similar to Random Forest but introduces stronger randomisation in split selection, which can improve robustness. Histogram Gradient Boosting is a tree boosting method optimised for structured numerical data. XGBoost is a regularised gradient boosting algorithm that sequentially improves weak learners.

The general ensemble prediction can be written as:

Table 2 Sequential expanding-window forecasting protocol

Forecast period	Information used	Forecast horizon
January 2021	Observed history up to December 2020	One month ahead
February–December 2021	Observed history updated monthly	One month ahead
2022–2024	Expanding history updated before each forecast month	One month ahead
January–October 2025	Observed history up to the month preceding each forecast	One month ahead

$$\hat{y}_i = \sum_{k=1}^{K} f_k(X_i) \tag{14}$$

where f_k is the k-th regression tree, K is the total number of trees, and X_i is the feature vector. For boosting models, the objective function can be expressed as:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{15}$$

where $l(y_i, \hat{y}_i)$ is the loss function and $\Omega(f_k)$ is the regularisation term used to control model complexity.

3.9 Evaluation metrics

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Coefficient of Determination (R^2), and Mean Absolute Percentage Error (MAPE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |P_i - \hat{P}_i| \tag{16}$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (P_i - \hat{P}_i)^2} \tag{17}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (P_i - \hat{P}_i)^2}{\sum_{i=1}^{n} (P_i - \bar{P})^2} \tag{18}$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{P_i - \hat{P}_i}{P_i} \right| \tag{19}$$

where P_i is the actual modal price, $\hat{P}_i$ is the predicted modal price, $\bar{P}$ is the mean actual price, and n is the number of test observations.

4 Results and discussion

4.1 Exploratory analysis

Table 1 presents the descriptive statistics and indicates substantial variation among commodities in terms of price level and volatility. Sesamum has the highest coefficient of variation, while Wheat and Cotton show comparatively lower coefficients of variation.

Figure 5 presents the coefficient of variation (CV) of modal prices across the selected agricultural commodities, where CV measures relative price variability by comparing the standard deviation with the mean price. A higher CV indicates greater relative dispersion, while a lower CV indicates comparatively smaller dispersion around the mean. The figure is used descriptively to summarise price variability across commodities and is not interpreted as a direct measure of out-of-sample forecastability. Agricultural price variability has also been documented in commodity price studies because prices are affected by seasonal supply, demand shocks, and market uncertainty [13, 14].

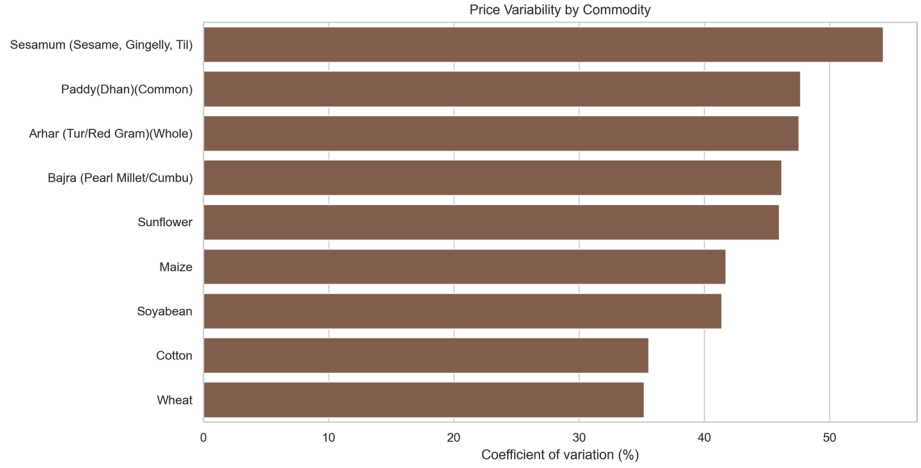


Fig. 5 Coefficient of variation by commodity

Table 3 Benchmark and best machine learning results for all commodities

Commodity	Benchmark	MAPE	R^2	Best ML	MAPE	R^2
Paddy	Naive	2.30	0.90	RandomForest	4.58	0.60
Wheat	Naive	2.46	0.94	RandomForest	5.45	0.73
Soyabean	Naive	3.05	0.90	XGBoost	8.60	0.47
Bajra	Naive	3.40	0.94	RandomForest	5.59	0.81
Cotton	Naive	3.84	0.82	RandomForest	10.41	− 0.15
Sesamum	Naive	4.60	0.91	ExtraTrees	11.74	0.24
Maize	Naive	4.62	0.83	RandomForest	7.07	0.57
Arhar	Naive	4.74	0.91	XGBoost	14.18	0.10
Sunflower	Naive	4.91	0.71	XGBoost	10.71	− 0.51

4.2 Benchmark evaluation

The leakage-safe analysis provides a benchmark-based evaluation of the empirical results. Table 3 reports the best benchmark and the best machine learning model for all nine commodities during January 2021–October 2025. The one-month Naive benchmark achieved the lowest MAPE for every commodity, with MAPE ranging from 2.30% for Paddy to 4.91% for Sunflower. This indicates that monthly agricultural prices in the test period showed strong short-run persistence, so the previous month price was a difficult benchmark to outperform.

Although the machine learning models did not outperform the one-month Naive benchmark, they performed better than the Seasonal Naive benchmark for all commodities. The complete 9 commodities × 6 methods results are provided as Supplementary Table S1. This distinction is important: the machine learning models captured useful nonlinear lag and rolling-window information relative to annual seasonal repetition, but they did not add enough information to improve on short-run price persistence.

Figure 6 shows that Seasonal Naive forecasts produced substantially higher errors than both Naive and the best machine learning models. The result suggests that price continuation from the immediately preceding month was more informative during 2021–2025 than the same calendar month from the previous year. This is consistent with short-term market persistence and gradual month-to-month adjustment in aggregated monthly prices.

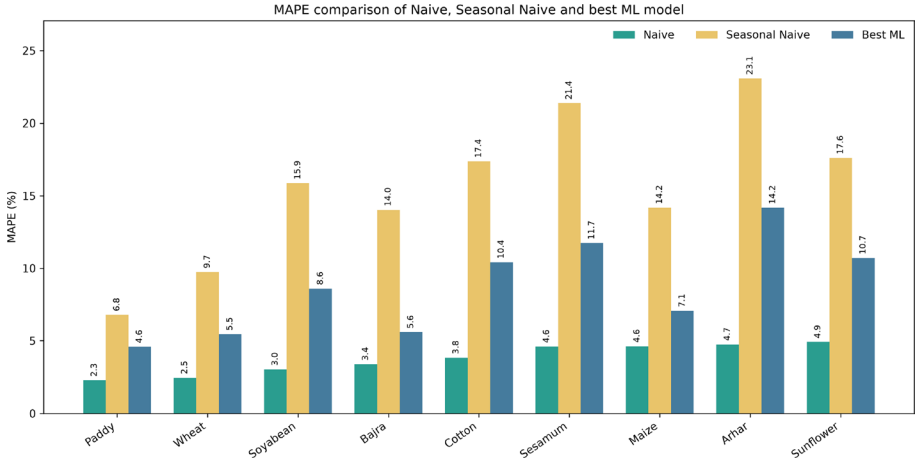


Fig. 6 MAPE comparison of Naive, Seasonal Naive, and best machine learning models



Fig. 7 Actual, Naive, and best machine learning forecasts for representative commodities

Figure 7 compares actual prices with Naive and best machine learning predictions for representative commodities. The plots show that machine learning forecasts generally follow broad price movements, but the Naive forecast is often closer during smooth month-to-month changes. Larger deviations are visible around abrupt price shifts, especially in more volatile commodities. Because the present study uses only historical price features, sudden changes caused by arrivals, rainfall, policy interventions, input costs, or wider market shocks cannot be fully anticipated.

4.3 Fixed and expanding window comparison

A concise comparison of fixed-window and expanding-window machine learning results is shown in Table 4. Expanding retraining reduced MAPE relative to fixed ML training

Table 4 Fixed versus expanding-window best machine learning comparison

Commodity	Best ML	MAPE	Exp.best ML	Exp.MAPE	MAPE change
Arhar	Extra Trees	28.85	XGBoost	14.18	−14.67
Bajra	Random Forest	19.49	Random Forest	5.59	− 13.90
Cotton	Extra Trees	11.60	Random Forest	10.41	− 1.20
Maize	Random Forest	8.12	Random Forest	7.07	− 1.05
Paddy	Random Forest	11.76	Random Forest	4.58	− 7.18
Sesamum	Extra Trees	17.68	Extra Trees	11.74	− 5.95
Soyabean	Random Forest	13.58	XG Boost	8.60	− 4.99
Sunflower	XG Boost	16.71	XG Boost	10.71	− 6.00
Wheat	Random Forest	14.69	Random Forest	5.45	− 9.24

for all nine commodities, indicating improved forecasting performance when newly observed prices were incorporated. However, even after expanding retraining, the best ML models did not outperform the one-month Naive benchmark.

Start-year sensitivity, mean-versus-median aggregation robustness, detailed hyperparameters, and monthly reporting diagnostics are reported in the n Material to keep the main manuscript concise. The sensitivity results show that no single training start year is uniformly best across all commodities, supporting the conclusion that commodity-specific market behaviour matters.

4.4 Discussion

The results indicate that historical monthly prices exhibit strong one-month persistence. The one-month Naive benchmark therefore provides a strong reference for evaluating the machine learning models. This explains why the Naive benchmark performs so well across all commodities. In a one-step-ahead monthly setting, the previous month's price already contains much of the information needed to forecast the next month when price changes are gradual.

The machine learning models still provide useful evidence because they consistently improve over the Seasonal Naive benchmark. This suggests that the engineered lag, rolling mean, rolling volatility, and calendar features capture information beyond simple annual seasonality. The improvement from fixed to expanding validation shows that periodic updating reduced ML forecast errors relative to a fixed ML model.

Commodity-wise differences remain important. Paddy and Wheat showed relatively low errors, whereas Arhar, Cotton, Sesamum, Soyabean, and Sunflower were more challenging for machine learning models during the test period. These differences may reflect variation in market volatility, reporting density, commodity-specific shocks, and the limited information available in historical-price-only features. The results therefore caution against claiming a single universally best machine learning model for all commodities.

External variables such as rainfall, temperature, market arrivals, minimum support price, fuel and transport costs, inflation, export–import policy, and festival demand were not used in the present models. These factors may explain sudden price movements that cannot be anticipated from historical prices alone. Their integration is therefore treated as a limitation and future research direction rather than as part of the current empirical design.

Deep learning models such as LSTM and GRU were not included because the present study focused on validating the forecasting pipeline, preventing leakage, and comparing

machine learning models with proper benchmarks. A rigorous recurrent-model comparison would require a separate chronological architecture-selection and tuning study. Future work may examine deep learning, probabilistic forecasting, and exogenous variables under the same leakage-safe evaluation principles.

5 Conclusion

This study developed and evaluated a machine learning framework for forecasting monthly agricultural commodity prices using long-term mandi-level price data for nine commodities in Maharashtra. The analysis used leakage-safe preprocessing, monthly aggregation, lag-based and rolling-window features, sequential one-step-ahead forecasting, and expanding-window validation for the January 2021–October 2025 test period.

The results show that the one-month Naive benchmark achieved the lowest MAPE for all nine commodities, with errors ranging from 2.30% for Paddy to 4.91% for Sunflower. The best machine learning models improved over the Seasonal Naive benchmark, and expanding-window retraining improved performance relative to fixed-window training, but the machine learning models did not outperform the one-month Naive benchmark. Therefore, the results support a more cautious conclusion: historical-price-only machine learning models can capture some temporal structure in monthly commodity prices, but short-run price persistence remains a very strong baseline.

The study provides a reproducible leakage-safe framework for evaluating historical-price-based agricultural price forecasts. Its main limitation is the absence of exogenous drivers such as rainfall, arrivals, production, input costs, policy changes, and wider market indicators. Future research should integrate such variables and examine whether richer information sources can improve on the strong Naive benchmark observed in this study.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s44564-026-00015-0.

Supplementary file 1

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Author contributions
All authors contributed equally to this work.

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Data availability
The data used in this study are publicly available through the AGMARKNET portal (https://agmarknet.gov.in)

Code availability
Code is available upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The authors declare no conflict of interest.

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References

1. Sari M, Duran S, Kutlu H, Guloglu B, Atik Z. Various optimized machine learning techniques to predict agricultural commodity prices. Neural Comput Appl. 2024;36(19):11439–59. https://doi.org/10.1007/s00521-024-09679-x.
2. Sun F, Meng X, Zhang Y, Wang Y, Jiang H, Liu P. Agricultural product price forecasting methods: a review. Agriculture. 2023;13(9):1671. https://doi.org/10.3390/agriculture13091671.
3. Oktoviany P, Knobloch R, Korn R. A machine learning-based price state prediction model for agricultural commodities using external factors. Decis Econ Finance. 2021;44:1063–85. https://doi.org/10.1007/s10203-021-00354-7.
4. Breiman L. Random forests. Mach Learn. 2001;45:5–32. https://doi.org/10.1023/A:1010933404324.
5. Friedman JH. Greedy function approximation: a gradient boosting machine. Ann Stat. 2001;29(5):1189–232. https://doi.org/10.1214/aos/1013203451.
6. Chen T, Guestrin C. XGBoost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining; 2016;785–794. https://doi.org/10.1145/2939672.2939785.
7. Geurts P, Ernst D, Wehenkel L. Extremely randomized trees. Mach Learn. 2006;63:3–42. https://doi.org/10.1007/s10994-006-6226-1.
8. Mahale Y, Khan N, Kulkarni K, Wagle SA, Pareek P, Kotecha K, et al. Crop recommendation and forecasting system for Maharashtra using machine learning with LSTM: a novel expectation-maximization technique. Discov Sustain. 2024;5:134. https://doi.org/10.1007/s43621-024-00292-5.
9. Nagesh OS, Budaraju RR, Kulkarni SS, et al. Boosting enabled efficient machine learning technique for accurate prediction of crop yield towards precision agriculture. Discov Sustain. 2024;5:78. https://doi.org/10.1007/s43621-024-00254-x.
10. Tran NQ, Felipe A, Nguyen T. Predicting agricultural commodities prices with machine learning: a review of current research. 2023. arXiv:2310.18646 [Preprint].
11. Hyndman RJ, Athanasopoulos G. Forecasting: principles and practice. 3rd ed. Melbourne: OTexts; 2021. https://otexts.com/fpp3/.
12. AGMARKNET. Agricultural Marketing Information Network (AGMARKNET). Directorate of Marketing and Inspection, Government of India. https://agmarknet.gov.in/home. Accessed January 2026.
13. Deaton A, Laroque G. On the behaviour of commodity prices. Rev Econ Stud. 1992;59(1):1–23. https://doi.org/10.2307/2297923.
14. Cashin P, McDermott CJ. The long-run behaviour of commodity prices: small trends and big variability. IMF Staff Pap. 2002;49(2):175–99. https://doi.org/10.2307/3872482.

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